PropRent

Property Rental Management & Tenant Portal Platform

Chapter 2: Literature Review & Comparative Analysis

Version 1.0

Chapter Author	Asmaa Hany (No-Code System Lead)
Project Team	1-Ahmed Fouda (No-Code Application Lead) 2-Asmaa Hany (No-Code System Lead) 3-Saad Walid (No-Code System Lead) 4-Yara Mokhtar (Delivery Lead) 5-Manar Gomaa (Integrations Lead) 6-Hager Saad (UX & Discovery Lead)
DEPI Deadline	16 October 2026
Review Scope	Existing rental-management platforms, supporting standards and a gap analysis against the Chapter 3 baseline (7 epics, 20 functional requirements, 10 non-functional requirements).

Table of Contents

1. Introduction & Review Method **3**

1.1 Purpose and Scope 3

1.2 Research Questions 3

1.3 Review Approach & Sources 3

2. Domain Background **3**

2.1 The Rental Management Lifecycle 3

2.2 Why Digitisation Matters 4

3. Review of Existing Platforms **4**

3.1 Selection Criteria 4

3.2 Platform Profiles 4

3.3 Cross-Platform Observations 5

4. Technology, Standards & Practice Review **5**

4.1 Payments 6

4.2 Electronic Signature and Legal Records 6

4.3 Privacy, Security and Accessibility 6

5. Comparative & Gap Analysis **6**

5.1 Comparison Criteria 6

5.2 Capability Coverage Comparison 7

5.3 Gap Analysis 8

5.4 PropRent Positioning 8

5.5 Limitations of the Review 8

5.6 Supervisor Evaluation, Feedback & Grading Criteria Alignment 8

6. Conclusions & Implications **9**

6.1 Key Findings 9

6.2 Implications for Subsequent Chapters 10

7. References & Standards Consulted **10**

> **Purpose of this chapter**
>
> This chapter reviews the platforms, standards and practices that surround PropRent. It identifies what established products provide, where gaps remain, and how the Chapter 3 baseline responds to them, so that design, implementation and testing rest on a documented foundation.

1. Introduction & Review Method

1.1 Purpose and Scope

This chapter reviews the knowledge and market context behind PropRent, a property rental management and tenant portal platform. It examines how established platforms support tenants, managers, owners and contractors, which technical and regulatory standards constrain a system that handles payments, leases and personal data, and where gaps remain that justify the PropRent baseline.

The chapter produces three outputs that feed the later chapters:

- a **capability landscape** of commercial rental-management platforms (Section 3);
- a **standards and practice review** covering payments, e-signature, privacy, security, accessibility and requirements engineering (Section 4);
- a **gap analysis** that maps each finding to the Chapter 3 functional and non-functional requirements (Section 5).

1.2 Research Questions

ID	Research Question	Answered In
RQ1	Which capabilities do established rental-management platforms typically provide to tenants, managers and owners?	Section 3
RQ2	Which technical and regulatory standards constrain payments, e-signature, privacy, accessibility and security?	Section 4
RQ3	Where do existing solutions leave gaps that matter to PropRent's target users?	Section 5.3
RQ4	How does the Chapter 3 baseline respond to those gaps while remaining a bounded graduation scope?	Sections 5.2 and 5.4

1.3 Review Approach & Sources

- Public product documentation and feature descriptions of commercial rental-management platforms.
- Official standards and legislation relevant to payments, signatures, data protection, security and accessibility.
- Requirements-engineering practice (MoSCoW prioritisation, Given-When-Then acceptance criteria, quality models).
- The PropRent Chapter 3 baseline: 8 stakeholders, 8 user stories, 3 use cases, 20 functional and 10 non-functional requirements.

> **Method note**
>
> Statements about commercial products are qualitative and based on publicly described capabilities. Features, pricing and regional availability change frequently, so they should be re-verified before the final presentation. No market-share or user-number claims are made in this chapter.

2. Domain Background

2.1 The Rental Management Lifecycle

Rental management is a chain of connected processes. Each stage involves a different mix of stakeholders, and the pain points recorded in the Chapter 3 stakeholder register concentrate where the stages hand over to one another.

Lifecycle Stage	Typical Activities	Recurring Pain Points
Marketing & Enquiry	Publish units with photos, answer enquiries, arrange viewings.	Outdated listings, slow replies, repeated form filling.
Lease & Signature	Generate lease from template, collect signatures, store documents.	Paper workflows, lost paperwork, template mapping errors.
Billing & Collection	Issue invoices, collect rent, apply late fees, issue receipts.	Missed due dates, opaque payment status, manual rent chasing.
Maintenance	Receive request, triage, assign contractor, verify completion.	Repairs reported by phone with no tracking, vague tickets, repeated site visits.
Renewal & Move-out	Propose new terms, re-sign, close or archive the lease.	Missed renewal windows, unclear owner approval.
Reporting	Occupancy, collections, arrears, owner statements.	Delayed or unclear statements, no audit trail.

2.2 Why Digitisation Matters

The stakeholder register describes spreadsheet-and-email workflows, duplicated data entry and limited visibility for owners as the main operational weaknesses. A single platform that links payments, maintenance and leases removes hand-offs between tools and creates one auditable record of what happened and who approved it.

> **Key insight**
> Tenants and Property Managers are the two groups whose daily workflows intersect on every core process (payments, maintenance, leases). The review therefore evaluates every platform and standard against both groups at once, and flags any trade-off between them.

3. Review of Existing Platforms

3.1 Selection Criteria

- The product is widely documented and positioned for rental or property management.
- It combines a tenant-facing portal with manager-facing tools (not only a listing site).
- It supports online rent payment and a maintenance request workflow.

3.2 Platform Profiles

Four commercial platforms were profiled. The profiles are qualitative summaries intended to describe the capability landscape, not to rank products.

Platform	Typical Target Segment	Capabilities Commonly Described	Relevance to PropRent
AppFolio Property Manager	Residential property managers with growing portfolios.	Online payments, listings and leasing, maintenance requests, reporting, owner and tenant portals.	Closest in breadth to PropRent; subscription product with a market orientation that may not match local conditions.
Buildium	Small to mid-sized property managers and associations.	Accounting, online payments, leasing tools, maintenance requests, owner and tenant portals.	Strong accounting emphasis; PropRent keeps a narrower, bounded financial scope (invoicing, payments, receipts).

Platform	Typical Target Segment	Capabilities Commonly Described	Relevance to PropRent
TenantCloud	Independent landlords and small managers.	Listings, online rent collection, maintenance requests, tenant screening options.	Lightweight for small portfolios; fewer multi-role governance controls are typically expected.
Yardi Breeze	Smaller portfolios within the Yardi ecosystem.	Accounting, online payments, maintenance, marketing tools.	Vendor-ecosystem dependency; illustrates the lock-in risk PropRent addresses through modular design.

3.3 Cross-Platform Observations

Across the profiled platforms the following patterns were observed. They are stated as tendencies, not as universal facts, and should be re-verified per product.

- **Common core:** online rent payment, recurring billing, maintenance requests, listings and tenant portals are offered almost everywhere.
- **Variable depth:** owner approval thresholds, granular audit trails and signed-document immutability differ by product or pricing plan.
- **Market orientation:** payment rails, screening services and lease templates are typically designed around specific national markets.
- **Closed and priced per unit:** subscription pricing and closed source limit adaptation for a local, graduation-scale deployment.

4. Technology, Standards & Practice Review

PropRent handles money, legally binding documents and personal data. The following standards and practices shape how the Chapter 3 requirements are written and how they will be verified.

Area	Standard / Practice	What It Contributes	Applied In PropRent
Payments	PCI DSS (SAQ A scope), tokenisation, 3-D Secure, idempotent requests, signed webhooks	Keeps card data off merchant servers and prevents duplicate charges from retries.	FR-FIN-02, FR-FIN-03, NFR-SEC-03, US-07
E-signature & records	Egyptian Electronic Signature Law No. 15 of 2004; comparable frameworks (eIDAS, ESIGN)	Defines legal effect of electronic signatures; audit certificates support evidence.	FR-LEASE-02, FR-LEASE-03, UC-03
Data protection	Egyptian Personal Data Protection Law No. 151 of 2020; GDPR as reference	Lawful handling, minimisation, retention and data-subject rights for tenant data.	NFR-SEC-01, NFR-SEC-02, FR-AUTH-03
Application security	OWASP Top 10 (Broken Access Control ranks first in the 2021 edition)	Guides RBAC, object-level ownership checks and secure session handling.	FR-AUTH-02, FR-AUTH-03, NFR-SEC-02
Accessibility	WCAG 2.1 Level AA	Contrast, keyboard operability and touch-target rules for mobile-first users.	NFR-USA-01
Requirements practice	MoSCoW; Given-When-Then; ISO/IEC/IEEE 29148; ISO/IEC 25010	Prioritisation, testable acceptance criteria and a shared vocabulary for quality attributes.	Chapter 3 Sections 2-4

4.1 Payments

Using gateway-hosted fields keeps PropRent within PCI DSS SAQ A scope, because only a token and the last four digits are stored. Idempotency keys tied to the invoice ID, signature and timestamp verification on webhooks and duplicate-event detection address the double-charge and replay risks that payment providers describe as their main pain points.

4.2 Electronic Signature and Legal Records

Egyptian Law No. 15 of 2004 regulates electronic signatures and the authority that licenses signing services. A built-in or third-party signing module must therefore provide identity binding, a tamper-evident audit certificate and immutable storage of the signed lease, which is what FR-LEASE-02 and FR-LEASE-03 require.

> **Legal note**
>
> This chapter summarises standards for engineering purposes only and is not legal advice. Legal effect of lease clauses, e-signatures and data processing should be confirmed with qualified counsel before any real deployment.

4.3 Privacy, Security and Accessibility

- Data minimisation, configurable retention and audit logging support Law No. 151 of 2020 and GDPR-style expectations.
- Server-side RBAC with object-level checks answers the most common web-application weakness (insecure direct object references).
- WCAG 2.1 AA and a mobile-first layout match the Chapter 3 observation that most tenants use the portal on phones.

5. Comparative & Gap Analysis

5.1 Comparison Criteria

- **Functional coverage** of the core lifecycle stages.
- **Governance and audit:** approvals, role scoping, immutable records.
- **Local fit and cost:** suitability for a locally focused, bounded deployment.
- **Extensibility:** ability to change providers or rules without redevelopment.

5.2 Capability Coverage Comparison

Capability	Typical Commercial Platform	PropRent Baseline	Reference	Priority
Recurring invoicing	Common	Included	FR-FIN-01	Must
Online payment & receipts	Common	Included	FR-FIN-02	Must
Autopay, late fees, webhook reconciliation	Common	Included	FR-FIN-03	Should
Maintenance request with photo/video	Common	Included	FR-MAINT-01	Must
Ticket tracking, triage, emergency handling	Common	Included	FR-MAINT-02	Must
Work orders and owner approval above a threshold	Varies by product	Included	FR-MAINT-03	Must
Property, unit and listing publication	Common	Included	FR-PROP-01, 02	Must
Public search and enquiry	Common	Included	FR-PROP-03	Must
Lease templates and generation	Common	Included	FR-LEASE-01	Must
E-signature with audit certificate	Common	Included	FR-LEASE-02	Must
Renewal workflow, immutable signed documents	Varies by product	Included	FR-LEASE-03	Must
Role-based access and audit logging	Common	Six roles	FR-AUTH-01 to 03	Must
Notifications and messaging	Common	Included	FR-COMM-01, 02	Must / Should
Manager, owner and tenant dashboards	Common	Included	FR-RPT-01 to 03	Must / Could
SMS and push, bulk import, social login, screening, accounting sync, portal syndication	Often offered	Deferred (Won't Have)	Chapter 3, Sec. 3.9	Won't

The baseline covers every core lifecycle stage. The deferred items are real market features but are not essential to the three core use cases, so they are recorded to prevent scope creep rather than omitted silently.

5.3 Gap Analysis

ID	Observed Gap	Impact	PropRent Response
G1	Tenant journey is fragmented across tools and channels (pay, report, sign).	Late fees, lost proof, chasing by phone.	Single mobile-friendly portal: US-01, UC-01, UC-02.
G2	Owners have limited control over repair spending.	Unauthorised or overpriced repairs.	Owner approval above a configurable threshold: FR-MAINT-03, US-03.
G3	Payment retries and gateway delays can create duplicates or mismatches.	Financial disputes and ledger errors.	Idempotency keys, webhook verification, reconciliation: FR-FIN-02, FR-FIN-03, US-07.
G4	Audit depth and signed-document immutability vary by product.	Weak evidence in disputes and reviews.	Append-only audit log and immutable signed leases: FR-AUTH-03, FR-LEASE-03, US-08.
G5	Commercial products assume specific markets and are priced per unit.	Poor local fit, cost and lock-in.	Bounded scope, configurable rules, provider-agnostic modules: NFR-MNT-01; local laws reviewed in Section 4.
G6	Accessibility and mobile usability are uneven.	Exclusion of users with disabilities and of phone-first tenants.	WCAG 2.1 AA, mobile-first design, 4-click payment target: NFR-USA-01, NFR-USA-02.

5.4 PropRent Positioning

PropRent does not try to outgrow established products in breadth. Its position is a focused, auditable platform for the full tenant-manager-owner-contractor loop, built to a verifiable baseline. Three points define it:

- **One loop, four roles:** payments, maintenance and leases share one record and one notification channel.
- **Governance by design:** approval thresholds, role and data scoping, and immutable signed documents are Must Have items.
- **Measurable quality:** ten NFRs carry explicit targets and verification methods (load testing, authorisation tests, accessibility audit, restore drill).

5.5 Limitations of the Review

- Commercial products were reviewed from public descriptions only; no hands-on trials or interviews were performed.
- Features and pricing change, so the comparison is a snapshot and must be re-verified.
- Legal references are summarised for engineering use and have not been reviewed by counsel.

5.6 Supervisor Evaluation, Feedback & Grading Criteria Alignment

The DEPI guidelines ask the Literature Review to document the lecturer's feedback and evaluation, suggested improvements and the final grading criteria. This section records them and will be updated as feedback arrives.

Feedback & Evaluation

Feedback Item	Action Taken	Where Reflected
Narrow the requirements scope to a bounded, verifiable baseline.	Stakeholders consolidated from 15 to 8, user stories from 11 to 8, functional requirements from 108 to 20 and NFRs from 60 to 10.	Chapter 3 v1.0, Sections 1, 2 and 3.9
Lecturer's formal assessment of this chapter.	Pending: to be recorded after the supervisor reviews this version.	This table

Suggested Improvements (proposed by the team)

Improvement	Rationale	Target
Re-verify platform capabilities before the final presentation.	Product features change.	Final presentation
Validate gap findings through stakeholder interviews and UAT sign-off.	Review so far relies on public sources.	Chapter 4 and testing
Review e-signature and data-protection clauses with legal counsel.	Standards were summarised for engineering use.	Before deployment
Run usability testing with at least 8 participants.	Verifies NFR-USA-02.	Testing phase
Reconsider deferred Won't Have items (e.g., SMS and push, bulk import) after the first release.	Marketing-ready features outside the MVP.	Future release

Final Grading Criteria Alignment

Grading Criterion	PropRent Evidence	DEPI Due Date
Documentation	Project planning, Literature Review and Requirements Gathering (this chapter and Chapter 3).	16 Oct 2026
Documentation	System Analysis & Design (diagrams, ERD, architecture, UI/UX).	6 Nov 2026
Implementation	Source code, GitHub repository, README, deployed application.	30 Nov 2026
Testing	Test plan, test cases, automated tests, bug reports.	4 Dec 2026
Presentation	User manual, technical documentation, presentation, infographic, optional demo video.	4 Dec 2026

> **Note on marks**
> The guidelines available to the team list the grading categories (documentation, implementation, testing, presentation) but not the mark weights. Update this table when the lecturer publishes them. All documents are to be uploaded to GitHub.

6. Conclusions & Implications

6.1 Key Findings

- The market already offers the core capabilities (payments, maintenance, leases, portals), so PropRent's value lies in a unified, auditable loop rather than novel features.
- The largest differences between products are governance depth, immutability of signed records and local fit.
- Payments, e-signature and personal data carry the strictest external constraints and were prioritised as Must Have items.
- The Chapter 3 baseline covers all core lifecycle stages with 20 functional and 10 non-functional requirements.

6.2 Implications for Subsequent Chapters

Chapter	What It Takes From This Review
Chapter 3: Requirements Gathering	Confirms the baseline scope, priorities and the deferred (Won't Have) list.
Chapter 4: System Analysis & Design	Uses the gap analysis for the problem statement and objectives; standards inform architecture, security and data design.
Testing & Quality Assurance	Standards supply verification methods: SAQ A self-assessment, authorisation tests, WCAG audit, restore drill.

> **Closing note**
> The findings in this chapter support, but do not replace, the Chapter 3 baseline. Any change to that baseline shall follow the project's formal change-control procedure.

7. References & Standards Consulted

- PCI Security Standards Council. *Payment Card Industry Data Security Standard (PCI DSS)* and SAQ A guidance.
- W3C. *Web Content Accessibility Guidelines (WCAG) 2.1*.
- OWASP Foundation. *OWASP Top 10 (2021)*.
- Arab Republic of Egypt. *Law No. 15 of 2004 regulating Electronic Signature*.
- Arab Republic of Egypt. *Personal Data Protection Law No. 151 of 2020*.
- European Union. *General Data Protection Regulation (EU) 2016/679*.
- ISO/IEC 25010. *Systems and software quality models*.
- ISO/IEC/IEEE 29148. *Requirements engineering*.
- Public product documentation of AppFolio, Buildium, TenantCloud and Yardi Breeze (to be re-verified before final submission).
- PropRent Chapter 3: *Requirements Gathering Specification*, v1.0.